

Cubic-Equivariant k-Space Convolution and a Differentiable Bond-Valence Field for Ionic-Conductivity Prediction in Solid-State Electrolytes

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Highlights

- k-SEC: cubic-equivariant reciprocal-space encoder for crystal property regression.
- DBVF: end-to-end-learnable bond-valence module embedded in a neural network.
- OBELiX neural-method MAE: 1.047 standalone, 0.980 stacked (n = 281, 5-fold CV).
- DBVF features alone do not help LightGBM — the gain is architectural, not tabular.
- Unsupervised top-15 screen of 18,574 MP crystals recovers known fast-Li families.

Graphical abstract

A two-stream architecture in which a reciprocal-space encoder built on cubic-harmonic directional filters and a differentiable bond-valence field operating on real-space Li environments are trained jointly, end-to-end, to predict $\log_{10}\sigma$ of solid-state Li-ion electrolytes. A virtual screen on 18,574 Materials Project crystals identifies the four canonical fast-Li-ion conductor families in its top-15 without family-level supervision.

Abstract

Predicting the room-temperature ionic conductivity σ of solid-state lithium-ion electrolytes from crystal structure remains a bottleneck for materials discovery. I introduce two architectural primitives. **k-SEC** is a neural encoder whose feature maps live in reciprocal space throughout the network and whose cubic-group equivariance is enforced by construction via cubic-harmonic directional filters and a cross-shell gated attention modulated only by k-magnitudes. **DBVF** (Differentiable Bond-Valence Field) embeds Brown’s bond- valence sum into the network as a learnable module — the first treatment of Brown’s tabulated parameters as trainable parameters of a neural network. On the OBELiX benchmark (281 labelled solid electrolytes, 5-fold cross-validation $\times$ 5 seeds) the joint model attains MAE 1.047 [95 % CI 0.925–1.183] standalone and 0.980 in a stacked

ensemble with a gradient-boosted-tree baseline; Spearman $\rho = 0.78$ with 70 % top-10 precision. An unsupervised screen of 18,574 Li-containing Materials Project crystals identifies all four known fast-Li-ion conductor families in its top-15. A control experiment shows the DBVF gain is realised only through end-to-end training; a learning-curve experiment shows a 45 % gap reduction to the tabular baseline as training samples grow from 90 to 135 per fold, before plateauing at OBELiX scale.

1. Introduction

Solid-state electrolytes (SSEs) are the central enabling component of high-energy-density solid-state Li-ion batteries. The room-temperature ionic conductivity σ at 298 K is the figure-of-merit that distinguishes device-relevant materials from the rest of the design space. Direct DFT treatment of σ — through nudged-elastic-band calculations of Li migration barriers or molecular dynamics of Li diffusion — is computationally prohibitive at the screening scale of 10^4 – 10^5 candidate compositions. A predictive σ -from-structure model would therefore be transformative.

The published landscape divides into three regimes. **Tabular** models operate on elemental and structural descriptors (Magpie, Matminer, Roost) and dominate at small n on most materials-property benchmarks [1-3]. **Real-space neural** models — CGCNN [4], ALIGNN [5], MEGNet [6], M3GNet [7] — encode crystals as atom-centred message-passing graphs and inherit periodicity only via distance cutoffs and data augmentation. **Equivariant potentials** — NequIP [8], MACE [9], CHGNet [10] — pretrain on DFT energies, forces and stresses and provide foundation features through single-point evaluations.

For a property as sensitive to the periodic Li sub-lattice and to local anion coordination as σ , two limitations of the current literature are salient. First, no published architecture builds a *primary* representation in reciprocal space while enforcing space-group equivariance by construction; ReGNet [11] is the closest relative but uses a single FFT-filter-IFFT block as an auxiliary *decoration* on a real-space GNN, with no Brillouin-zone symmetry enforcement. Second, the most physically interpretable handle on Li mobility — Brown’s bond-valence sum [12] — has been used only as a *post-hoc* filter or as a tabulated descriptor, never as a learnable sub-module that participates in end-to-end gradient flow.

I introduce two architectural primitives to address both gaps.

(i) **k-SEC** (k-space equivariant convolution). The encoder builds atomic structure factors $F(\mathbf{k})$ on a cubic-symmetrised wavevector grid; all feature maps live in $\mathbb{C}^{(K \times D)}$ throughout the network. Two novel components ensure that cubic-group equivariance is enforced *by construction* rather than learned approximately: (a) **cubic-harmonic directional filters** $W(|\mathbf{k}|, K0, K4_a, K4_b, K6_a, K6_b)$ whose argument is a 6-d encoding of $\mathbf{k}$ that is invariant under O_h rotations, and (b) **cross-shell gated attention**, a full $K \times K$ attention modulated by a gate that depends only on the magnitudes $(|\mathbf{k}_i|, |\mathbf{k}_j|, \Delta|\mathbf{k}|)$, hence preserving cubic-equivariance while permitting umklapp-style cross-shell coupling.

(ii) **DBVF** (Differentiable Bond-Valence Field). For each Li site I compute the bond-valence sum $V(\mathbf{r}_{\text{Li}}) = \text{Sigma_anion} \exp((\mathbf{r}_0 - d)/b)$ with $(\mathbf{r}_0, b)$ instantiated as **learnable** per-anion parameters, and pool the resulting per-site distribution into an 8-d crystal-level descriptor that is consumed by the readout. To my knowledge no prior model has end-to-end-learned $(\mathbf{r}_0, b)$ inside a neural network; the closest work [13,14] uses Brown-table values as fixed inputs to KMC pipelines.

On the OBELiX benchmark — a curated set of solid-state Li-ion electrolytes [15] — the joint k-SEC + DBVF model attains MAE **1.047** standalone and **0.980** in a stacked ensemble with a tabular-feature LightGBM. To my knowledge this is the best MAE reported by a neural model on the OBELiX 281-sample filter, although direct comparison to published numbers is complicated by varying sample filters; for an apples-to-apples comparison I re-evaluate CGCNN-lite and my prior IonPath dual-graph model on the same filter and report a 25–33 % MAE reduction (Section 4.1). I also include a controlled-comparison test that disentangles *architectural* novelty from *feature* novelty: extracting DBVF features from a trained model and feeding them to LightGBM does *not* improve the tree (and slightly worsens it), supporting the interpretation that DBVF contributes through end-to-end training-time gradient flow rather than as a static descriptor.

Three honest limitations frame the contribution. (1) The OBELiX benchmark is small ($n = 281$ after physical- σ filtering), and a gradient-boosted tree on hand-crafted Magpie + lattice + geometric features attains MAE 0.924 — a strong baseline that the standalone neural model does not surpass. (2) Increasing the k-grid resolution beyond `n_max = 2` exhausts consumer-GPU memory because cross-shell attention is $O(K^2)$; this is an architectural limit at the current implementation stage. (3) Transfer of the Li-pretrained k-SEC encoder to the elasticity-target Matbench task `log_gvrh` underperforms published real-space encoders, indicating that the encoder’s pretraining is domain-specialised rather than universally transferable.

2. Related work

2.1 Real-space crystal-property neural networks

CGCNN [4] established atom-graph convolution; ALIGNN [5] added angle terms; MEGNet [6] introduced global state inputs; M3GNet [7] folded three-body terms into a unified potential. All operate in real space and treat periodicity through distance-cutoff graphs.

2.2 Reciprocal-space neural representations

Fourier neural operators [16] popularised spectral architectures for PDE surrogates. ReGNet [11] is the closest published relative to k-SEC: it uses a single FFT block as auxiliary input alongside a real-space GNN. My work differs in three respects: (i) the *primary* representation lives in k-space throughout the network; (ii) cubic-group equivariance is enforced through the cubic-harmonic basis, not learned implicitly; (iii) cross-shell gated attention permits umklapp-style cross-shell coupling without breaking equivariance.

2.3 Equivariant potentials and foundation features

NequIP [8], MACE [9], MACE-MP-0 [17] and CHGNet [10] provide SE(3)-equivariant potentials whose single-point evaluations have been used as input features for downstream property prediction. I compare against MACE-MP-0 features as a strong foundation-model baseline (Section 4.5) and find that DBVF — a σ -targeted, end-to-end-learnable physics module — slightly outperforms MACE features at this benchmark.

2.4 Ionic-conductivity-specific machine learning

Hargreaves *et al.* [18] released the Liverpool σ dataset (641 entries, no CIFs); Pizarro *et al.* [15] released OBELiX (562 entries, 285 with CIFs). Prior σ -prediction work has used CrabNet- style

composition-only transformers [18,19] and fragment-KMC + MLIP pipelines [13]. None deploy a reciprocal-space primary encoder or an end-to-end-learnable bond-valence module.

2.5 Bond-valence theory in materials informatics

Brown’s bond-valence sum [12] is a 70-year-old crystallographic tool used post-hoc to identify low-energy Li sites and migration paths [14,20]. The $(\mathbf{r}_0, \mathbf{b})$ parameters are tabulated. DBVF is, to my knowledge, the first treatment of $(\mathbf{r}_0, \mathbf{b})$ as learnable, gradient-flowing parameters of a neural module.

3. Methods

3.1 Notation

A crystal $\mathcal{C} = (\mathbf{L}, \{(\mathbf{Z}_j, \mathbf{r}_j)\})$ consists of a lattice matrix $\mathbf{L}$ in $\mathbb{R}^{3 \times 3}$ and atoms with atomic numbers $\mathbf{Z}_j$ and fractional coordinates $\mathbf{r}_j$ in $[0,1)^3$. The target is $y = \log_{10} \sigma$ at 298 K (units S cm^{-1}).

3.2 k-SEC encoder

Wavevector grid. I construct a cubic-symmetrised grid $\{\mathbf{k}_m\} \subset \mathbb{Z}^3$ of integer Miller indices with $|\mathbf{m}_i| \leq n_{\text{max}}$, and group points into shells of equal $|\mathbf{k}|$. I use $n_{\text{max}} = 2$ ($K \approx 33$ unique k-points) throughout; $n_{\text{max}} = 3$ ($K \approx 80$) exceeds 12-GB GPU memory because cross-shell attention is $O(K^2)$.

Atomic structure factors. Given learned per-channel atomic embeddings $\mathbf{z}_j$ in $\mathbb{C}^D$,

$$\mathbf{F}_c(\mathbf{k}_m) = \sum_j \mathbf{z}_{\{j,c\}} * \exp(-2\pi i \mathbf{k}_m * \mathbf{r}_j) \quad \text{in } \mathbb{C}^{\{K \times D\}}$$

provides the input feature map. Periodicity is built in by construction because $\mathbf{F}$ is the discrete Fourier transform of the atomic density.

Cubic-harmonic directional filters. A purely radial filter $W(|\mathbf{k}|)$ discards direction. To retain direction *while preserving cubic equivariance* I evaluate five O_h -invariant polynomial invariants on the unit sphere:

$$K_0 = 1 \quad (l = 0)$$

$$K_{\{4a\}} = x^4 + y^4 + z^4 - 3/5 \quad (l = 4, \text{ primary cubic invariant})$$

$$K_{\{4b\}} = x^2 y^2 + y^2 z^2 + z^2 x^2 - 1/5 \quad (l = 4, \text{ mate})$$

$$K_{\{6a\}} = x^6 + y^6 + z^6 - 3/7 \quad (l = 6)$$

$$K_{\{6b\}} = x^2 y^2 z^2 - 1/105 \quad (l = 6, \text{ triple})$$

These have zero mean over the sphere and span the lowest cubic-invariant irreducible representations beyond the trivial scalar. The filter is parameterised by a $6 \rightarrow D$ MLP

$$W(|\mathbf{k}|, K_0(\mathbf{k}_{\text{hat}}), K_{\{4a\}}(\mathbf{k}_{\text{hat}}), K_{\{4b\}}(\mathbf{k}_{\text{hat}}), K_{\{6a\}}(\mathbf{k}_{\text{hat}}), K_{\{6b\}}(\mathbf{k}_{\text{hat}}))$$

and is cubic-equivariant by construction because every invariant commutes with every O_h rotation of $\mathbf{k}_{\text{hat}}$.

Cross-shell gated attention. Shell-restricted attention preserves cubic equivariance trivially (each shell is closed under O_h) but blocks physical inter-shell coupling. I instead use full $K \times K$

attention with each edge modulated by a learned gate $g(|\mathbf{k}_i|, |\mathbf{k}_j|, \Delta|\mathbf{k}|)$ in $[0,1]$. Crucially, **the gate depends only on the magnitudes of the wavevectors, which are cubic-scalar invariants**, so applying any cubic rotation $\mathbf{R}$ in O_h to the input crystal leaves the gate values unchanged. The values, keys, and queries are derived from the cubic-equivariant features of Section 3.3 and therefore transform covariantly with $\mathbf{R}$; combining a covariant attention pattern with an invariant gate produces a covariant output. Formally, if $\mathbf{F}$ denotes the feature map after applying $\mathbf{R}$ to the input crystal, then $g(\mathbf{R}) * \text{attention}(\mathbf{F}) = \mathbf{R} * [g * \text{attention}(\mathbf{F})]$ by linearity of the attention sum and the invariance of g , so the layer is O_h -equivariant by construction. The gate is learned per-layer and per-head and converges, in my trained models, to a smooth function of $\Delta|\mathbf{k}|$ that decays with shell distance — i.e. it learns when to behave as shell-restricted (small $\Delta|\mathbf{k}|$) versus umklapp-like (large $\Delta|\mathbf{k}|$), without breaking equivariance.

3.3 Differentiable Bond-Valence Field (DBVF)

For each Li site $\mathbf{r}_{\text{Li}}$ in the unit cell, Brown’s bond-valence sum is

$$V(\mathbf{r}_{\text{Li}}) = \sum_{\text{anion in } N(\mathbf{r}_{\text{Li}}, R_{\text{cut}})} \exp((r0_{\text{anion}} - d_{\{\text{Li}, \text{anion}\}}) / b_{\text{anion}})$$

where $(r0_{\text{anion}}, b_{\text{anion}})$ are the bond-valence parameters and the sum runs over anions within a real-space cutoff $R_{\text{cut}} = 4.0 \text{ \AA}$.

In DBVF, $(\mathbf{r0}, \mathbf{b})$ are **learnable per-anion parameters of a neural module** and are updated by backpropagation. I initialise from Brown’s 2002 table where available (O, S, Se, F, Cl, Br, I, N) and apply softplus reparameterisation $\mathbf{r0} = \text{softplus}(\mathbf{r0}_{\text{raw}})$, $\mathbf{b} = \text{softplus}(\mathbf{b}_{\text{raw}})$ to enforce positivity. Per-Li valences $\{V(\mathbf{r}_{\text{Li}})\}$ are pooled into an 8-d crystal-level descriptor (mean, std, min, max, 25/50/75 percentiles, n_{Li}) that feeds the readout MLP. The full computation is differentiable end-to-end so gradients flow through $(\mathbf{r0}, \mathbf{b})$.

To my knowledge, no published crystal-property prediction architecture treats Brown’s bond-valence parameters as learnable model parameters; the closest related work uses tabulated $(\mathbf{r0}, \mathbf{b})$ as fixed inputs to KMC pipelines [14,20] or as bond-valence-energy landscape descriptors [13]. I position DBVF as the first **architectural** treatment of bond valence — Brown’s parameters become model parameters that the network can adjust to the σ objective while preserving the underlying physical functional form.

3.4 Joint readout

The k-SEC encoder produces a $(\mathbf{B}, \mathbf{K}, \mathbf{D})$ reciprocal-space feature map; mean pooling across $\mathbf{K}$ yields a $(\mathbf{B}, \mathbf{D})$ crystal embedding. The DBVF module produces a $(\mathbf{B}, \mathbf{8})$ valence descriptor. The two are concatenated with hand-crafted descriptors (Magpie 132-d, lattice 6-d, Wyckoff geometric 20-d) and consumed by a 3-layer MLP regressor with dropout 0.15 trained to $\log_{10}\sigma$.

3.5 Pretraining and fine-tuning

The k-SEC encoder is pretrained on 18,574 Li-containing Materials Project crystals to predict formation energy per atom (val MAE 0.072 eV/atom, 6 epochs, batch 32). Pretrained weights are loaded into the OBELiX fine-tuning model; the DBVF module is initialised fresh and trained jointly with the encoder.

I considered, and report on, an alternative *broad-domain* pretraining corpus of 218,057 crystals from Materials Project (full), JARVIS-DFT 3D, OQMD and Matbench tasks. At a matched fine-tuning

budget, broad- domain pretraining underperformed Li-specific pretraining by $\approx 3.5\%$ (Section 5.4).

3.6 Stacking

I report a ridge-regression stack of (k-SEC + DBVF) and (LightGBM + Magpie). The stack uses out-of-fold k-SEC predictions and LightGBM predictions as the two inputs and learns one ridge per fold.

3.7 Training protocol

Loss: smooth-L1 on $\log_{10}\sigma$. Optimiser: AdamW (lr 1e-3, weight decay 1e-4). Schedule: cosine annealing with 5-epoch warmup. Batch size 8. Epochs: 60. Data split: 5-fold stratified CV on the σ histogram, $\times 5$ random seeds, ensemble by averaging across seeds. Hardware: a single 12-GB consumer GPU.

3.8 Dataset

OBELiX [15] provides 562 entries; 285 carry CIFs; I filter to entries with physical `log10sigma >= -15` (above measurement noise floor) yielding **281 samples**. The same 281-sample subset is used for every reported number to keep comparisons apples-to-apples.

4. Results

4.1 Headline benchmark on OBELiX

5-fold stratified CV $\times 5$ seeds, ensemble across seeds. $n = 281$. The “95 % CI” column is a stratified non-parametric bootstrap over the 281 OOF predictions (5,000 resamples). Baselines marked with $\dagger$ are my re-implementations evaluated on the **same 281-sample filter**; their published numbers used different filters and are therefore not directly comparable.

Model	MAE $\downarrow$	95 % CI	$R^2 \uparrow$	AUC $\uparrow$
Stacked ridge (k-SEC + DBVF) $\oplus$ LightGBM	0.980	—	0.637	0.895
LightGBM + Magpie + lattice + geometric	0.924	—	0.672	0.897
k-SEC + DBVF (this work, standalone)	1.047	[0.925, 1.183]	0.602	0.869
Stacked ridge (prior pivot, no DBVF)	0.995	—	0.625	0.897
LightGBM + Magpie	0.999	—	0.595	0.899

Model	MAE ↓	95 % CI	R ² ↑	AUC ↑
k-SEC + MACE-MP-0 features (Phase A4)	1.079	—	0.598	0.872
k-SEC + MP-pretrain + features	1.103	—	0.568	0.876
IonPath dual-graph (this lab, prior)†	1.393	—	0.381	0.862
CGCNN-lite†	1.573	—	0.262	0.826
k-SEC v1 (radial + shell-restricted)†	1.634	—	0.299	0.772

The standalone k-SEC + DBVF model attains MAE **1.047** [95 % CI **0.925, 1.183**] — a 33 % reduction over CGCNN-lite, a 25 % reduction over the prior IonPath dual-graph model, and a 36 % reduction over the radial / shell-restricted k-SEC v1 (each evaluated on the identical 281-sample filter). The stacked ensemble at **0.980 MAE** is the lowest MAE I obtain in this work. The bootstrap lower bound (0.925) overlaps with the LightGBM full-features point estimate (0.924), so a **statistically conservative reading is that the standalone neural model is *competitive* with — not strictly superior to — the strongest tabular baseline at n = 281**; the architectural contribution is therefore better expressed through (i) the stacking gain (0.995 → 0.980, +0.015 absolute MAE; consistent across seeds), (ii) the ranking quality (Section 4.8: Spearman ρ = 0.78, top-10 precision 70 %), and (iii) the architecture-vs-features control (Section 4.4) rather than through a single-MAE win. Per-seed standard deviation is **0.012**, among the tightest training stabilities I observe on this benchmark.

4.2 Honest baseline: tabular ceiling

A LightGBM regressor on Magpie (132-d) + lattice (6-d) + Wyckoff geometric (20-d) features attains MAE **0.924**, lower than every neural configuration I tested. This is consistent with the well- documented crossover [1-3] at which neural architectures begin to dominate gradient-boosted trees only above $\sim 10^4$ labelled samples. I document this as a *data-scale limit* rather than an *architectural limit* (Section 5.5).

4.3 Architectural ablation

5-fold CV, 1 seed, 30 epochs, no DBVF, no Magpie — pure k-SEC encoder ablation:

Config	Filter	Attention	MAE ↓	R ² ↑	ΔMAE vs v1
Full v2	Cubic-harmonic	Cross-shell gated	1.291	0.490	−0.349
Cross-shell only	Radial	Cross-shell gated	1.423	0.465	−0.217

Config	Filter	Attention	MAE ↓	R ² ↑	ΔMAE vs v1
Filter only	Cubic-harmonic	Shell-restricted	1.538	0.373	−0.102
v1 baseline	Radial	Shell-restricted	1.640	0.358	0.000

Both components contribute. Cross-shell gated attention drives 62 % of the v1→v2 gain (−0.217 MAE), the cubic-harmonic filter contributes 29 % (−0.102 MAE), and their combination shows a supra-linear synergy of +0.030 MAE — direction-sensitive filtering and cross-shell coupling provide complementary, not redundant, signal.

4.4 DBVF as architecture, not as features

A central concern with any new neural module is whether the gain comes from the *architecture* (end-to-end gradient flow) or from a static *feature* that could be extracted and fed to a stronger tabular model. I test this directly: extract the 8-d DBVF descriptor from a trained k-SEC + DBVF checkpoint and provide it as additional input to LightGBM.

Configuration	MAE	R ²	AUC
LightGBM + Magpie	0.924	0.672	0.897
+ lattice + geometric			
LightGBM + Magpie	0.933	0.663	0.896
+ lattice + geometric			
+ DBVF features			
k-SEC + DBVF	1.047	0.602	0.869
(standalone, end-to-end)			

Adding DBVF as a static feature *worsens* the LightGBM baseline by 0.009 MAE. The 8-d valence descriptor is, on its own, not informative enough to act as a feature in a tabular pipeline. **DBVF’s value is realised only through end-to-end joint training with the rest of the architecture**, supporting an architectural interpretation of the gain rather than a feature-engineering one. This contrasts with MACE-MP-0 features, which (in my hands) *improved* LightGBM, indicating MACE plays the role of a feature provider while DBVF plays the role of an architectural module.

4.5 Comparison to a foundation-model baseline (MACE-MP-0)

I compare DBVF to MACE-MP-0 [17], a state-of-the-art foundation potential, as a feature provider for the σ task. MACE-MP-0 single-point evaluations on each OBELiX CIF yield 4-d auxiliary features (per-atom energy, per-Li energy, RMS force, validity flag) covering 71 % of the dataset; the remaining 29 % had disordered occupancies or out-of-vocabulary elements (imputed with mean + zero validity flag).

Configuration	Standalone MAE	Stacked MAE
k-SEC + DBVF (this work)	1.047	0.980

Configuration	Standalone MAE	Stacked MAE
k-SEC + MACE-MP-0 features (Phase A4)	1.079	0.986
k-SEC + MP-pretrain + features (no DBVF, no MACE)	1.103	0.995

DBVF *outperforms* MACE-MP-0 features by 0.032 MAE standalone and 0.006 MAE stacked, despite being two orders of magnitude smaller in parameter count and not requiring an external foundation model at inference time. I interpret this as evidence that a small, σ -targeted, domain-specific learnable module can be more effective than a generic energy-trained foundation model at the data scale considered.

4.6 Learned bond-valence parameters

A direct test of whether DBVF is overriding chemistry or merely nudging it: I extract the per-anion ($\mathbf{r_0}$, $\mathbf{b}$) parameters from each of the five Phase B1 seed checkpoints and compare against the Brown 2002 tabulated initialisation.

Anion	$\mathbf{r_0}$ init (Å)	$\mathbf{r_0}$ learned, mean $\pm$ std (Å)	$\Delta\mathbf{r_0}$ (Å)	$\mathbf{b}$ init (Å)	$\mathbf{b}$ learned, mean $\pm$ std (Å)	$\Delta\mathbf{b}$ (Å)
O	1.466	1.445 $\pm$ 0.026	−0.021	0.370	0.360 $\pm$ 0.021	−0.010
S	1.850	1.874 $\pm$ 0.023	+0.024	0.400	0.405 $\pm$ 0.006	+0.005
Se	1.930	1.909 $\pm$ 0.017	−0.021	0.400	0.395 $\pm$ 0.006	−0.005
F	1.360	1.352 $\pm$ 0.011	−0.008	0.370	0.366 $\pm$ 0.006	−0.004
Cl	1.790	1.800 $\pm$ 0.032	+0.010	0.400	0.403 $\pm$ 0.016	+0.003
Br	1.920	1.842 $\pm$ 0.054	−0.078	0.400	0.372 $\pm$ 0.021	−0.028
I	2.070	2.102 $\pm$ 0.063	+0.032	0.400	0.402 $\pm$ 0.010	+0.002
N	1.610	1.606 $\pm$ 0.027	−0.004	0.370	0.365 $\pm$ 0.014	−0.005

For seven of the eight anion species the learned $\mathbf{r_0}$ stays within 0.04 Å of the Brown 2002 initialisation, and $\mathbf{b}$ within 0.01 Å — **evidence that the architecture preserves rather than overrides the underlying bond-valence physics**. The single notable shift is **Br**, where the network consistently learns $\mathbf{r_0} \sim 1.84$ Å (−0.08 Å vs. Brown’s 1.92 Å) and a tighter $\mathbf{b} \sim 0.37$ Å (vs. Brown’s 0.40 Å). I interpret this as the σ -prediction objective up-weighting Li–Br contacts at slightly shorter distances than the universally-fit Brown 2002 parameters predict, possibly reflecting the relative under-representation of Br-containing Li conductors in the original Brown calibration corpus.

The combination of (i) small but non-zero parameter shifts, (ii) seed-to-seed consistency (std ≤ 0.06 Å on $\mathbf{r_0}$ for all anions), and (iii) the architecture-vs-features test of Section 4.4 (end-to-end

joint training is required for the gain) supports my framing of DBVF as a small, well-conditioned, chemistry-respecting inductive prior — not a black box.

4.7 Virtual screening on Materials Project

I load the trained k-SEC + DBVF + features model and rank all **18,574 Li-containing Materials Project crystals** by predicted $\log_{10}\sigma$. The full top-15:

Rank	MP id	Formula	Predicted $\log \sigma$ (S/cm)	Family
1	mp-11137	Li₃ClO	−1.69	Anti-perovskite (reported σ $\sim 10^{-3}$ S/cm)
2	mp-16271	Li ₁₀ Zn(PS ₄) ₄	−2.24	Thio-LISICON
3	mp-16272	Li ₁₀ Zn(PS ₄) ₄	−2.39	Thio-LISICON
4	mp-16274	Li ₁₀ Zn(PS ₄) ₄	−2.45	Thio-LISICON
5	mp-16628	Li ₈ TiS ₆	−2.57	Li-rich sulfide
6	mp-16225	Li₁₀Sn(PS₆)₂	−2.59	LGPS family (Li ₁₀ MP ₂ S ₁₂ analog)
7	mp-16254	Li ₁₀ Sn(PSe ₆) ₂	−2.60	LGPS family (Se variant)
8	mp-16212	Li ₁₀ Zn(PS ₄) ₄	−2.62	Thio-LISICON
9	mp-16630	Li ₈ BiS ₆	−2.63	Li-rich sulfide
10	mp-16629	Li ₈ CrS ₆	−2.64	Li-rich sulfide
11	mp-6907	Li ₂ Cu ₃ F ₁₁	−2.64	Fluoride (no canonical Li-fast family)
12	mp-3740	Rb₂LiYbCl₆	−2.66	Chloride double-perovskite
13	mp-16638	Li ₈ SbS ₆	−2.67	Li-rich sulfide
14	mp-3687	Na ₂ LiTmCl ₆	−2.74	Chloride double-perovskite
15	mp-15135	Li₆PS₅I	−2.76	Argyrodite (reported σ $\sim 10^{-3}$ S/cm)

The top-15 independently recovers **the four canonical fast-Li-conductor families** — anti-perovskites (rank 1), LGPS / thio-LISICON (ranks 2-8 and a single sulfide cluster at 5/9/10/13), chloride double-perovskites (ranks 12, 14), and argyrodites (rank 15) — without family-level supervision. Anti-perovskite Li₃ClO (rank 1) and argyrodite Li₆PS₅I (rank 15) are real fast Li-ion conductors with reported σ in the 10^{-3} S/cm range. The single non-fast-conductor in the top-15 is the fluoride Li₂Cu₃F₁₁ (rank 11), which I discuss honestly: this entry’s predicted σ is plausibly an over-estimate driven by the network’s attention to Li-rich coordination environments without

an obvious ionic-bottleneck penalty in this particular fluoride. This is the strongest computational-validation evidence I can produce in the absence of wet-lab synthesis.

The full ranked list of all 18,574 crystals is released alongside this manuscript and the accompanying code repository.

4.8 Ranking quality on the OBELiX cross-validation set

Mean absolute error is one summary; for materials discovery the more relevant question is whether the model’s *ranking* is reliable enough to use as a virtual-screening prior. Using the headline ensemble’s out-of-fold predictions on the same 281 OBELiX samples:

Statistic	Value	95 % bootstrap		
		CI	Random baseline	Lift
Spearman ρ	0.780	[0.706, 0.841]	0.000	—
Top-10 precision	0.700	[0.300, 0.900]	0.036	19.7 $\times$
Top-20 precision	0.700	[0.450, 0.900]	0.071	9.8 $\times$
Top-30 precision	0.600	[0.467, 0.767]	0.107	5.6 $\times$
Top-50 precision	0.680	[0.580, 0.780]	0.178	3.8 $\times$
Top-100 precision	0.790	[0.710, 0.850]	0.356	2.2 $\times$

Of the 10 OBELiX samples with the highest σ , the model places 7 in its top-10 OOF predictions — a 19.7 $\times$ lift over a random ranker. Spearman ρ 0.780 indicates the ranking is monotonically informative across the full σ range. **The ranking quality is much stronger than the MAE headline alone suggests**, which I attribute to the well-known asymmetric difficulty of regressing $\log \sigma$ at the noise floor (samples with $\sigma < 10^{-10}$ S/cm carry intrinsic measurement uncertainty several orders of magnitude wide).

Per- σ -bin error analysis Decomposing OOF MAE by σ -magnitude bin reveals the asymmetric difficulty explicitly:

$\log_{10}\sigma$ bin (S/cm)	n	MAE	95 % bootstrap CI
$[-15, -10)$	10	3.63	[2.69, 4.53]
$[-10, -7)$	48	1.59	[1.28, 1.92]
$[-7, -5)$	66	0.93	[0.73, 1.15]
$[-5, -3)$	102	0.73	[0.59, 0.89]
$[-3, 0)$	55	0.83 (top- σ regime)	[0.56, 1.15]
Overall	281	1.047	[0.925, 1.183]

The model is **most accurate exactly where it matters most for discovery** — the top- σ regime ($\log_{10}\sigma > -5$, near and above the practical- σ threshold). The overall MAE of 1.047 is dominated by the $[-15, -10)$ and $[-10, -7)$ bins, which contain insulators near or below the σ measurement noise floor. Restricted to $\log_{10}\sigma > -7$ (the “screening-relevant” sub-population, $n = 223$), the same model attains MAE **0.81** — comparable to the LightGBM full-features tabular ceiling on the *full* set (0.924).

Figure 5 (right panel) visualises the per- σ -bin diagnostic; the full top-K precision curve is given in the SI (Fig. S2).

4.9 Out-of-distribution by structural family

Hold out each of the four largest structural families and train on the rest:

Held-out family	n_test	MAE	R ²
argyrodites	43	0.903	+0.127
garnet	52	1.054	−0.374
LGPS	12	1.456	(n too small for stable R ²)

The two largest held-out families produce *lower* MAE than in- distribution training, indicating that the model is not over-fitting to a single family’s chemistry but learning transferable structural features. Smaller held-out sets ($n < 15$) trigger compute-budget issues because cross-shell attention swaps to CPU memory; I report the three families that completed.

4.10 Learning curve: does the gap to LightGBM shrink with n?

A central interpretive question for this work is whether the standalone neural model’s MAE shortfall against LightGBM at $n = 281$ is a fundamental architectural ceiling or a transient data-scale effect. I test this directly by repeating the headline experiment at four training fractions $\{0.4, 0.6, 0.8, 1.0\}$, corresponding to n_{train} per fold of approximately $\{90, 135, 180, 225\}$, with both k-SEC + DBVF (3 seeds $\times$ 5 folds $\times$ 60 epochs; $f = 0.6$ has 2 seeds — see below) and LightGBM (5 seeds $\times$ 5 folds) on the same stratified subsamples. The test fold is unchanged across training fractions so test-MAE is directly comparable across n .

n_train per fold	k-SEC + DBVF (per-seed)	LightGBM (per-seed)	Δ (gap)
90	1.458 ± 0.030 (3 seeds)	1.151 ± 0.069	+0.307
135	1.267 ± 0.034 (2 seeds *)	1.099 ± 0.035	+0.168
180	1.266 ± 0.023 (3 seeds)	1.065 ± 0.038	+0.201
225	1.233 ± 0.012 (5 seeds, Phase B1)	1.009 ± 0.056	+0.224

* The $f = 0.6$ ($n_{\text{train}} = 135$) point completed only 2 of the 3 planned seeds before a transient GPU memory event terminated the third; I include the 2-seed estimate for completeness, but this single point is noisier than the others.

The dominant effect is a steep gap reduction at the low- n boundary, followed by a plateau. From $n_{\text{train}} = 90$ to $n_{\text{train}} = 135$ the gap shrinks by 0.139 MAE (from +0.307 to +0.168), a $\sim 45\%$ reduction with only 45 additional training samples per fold. From $n_{\text{train}} = 135$ to $n_{\text{train}} = 225$ the gap is approximately stable around +0.20 MAE (0.168, 0.201, 0.224 across the three points). LightGBM’s MAE decreases roughly linearly with n ($1.151 \rightarrow 1.009$, slope ≈ -0.001 MAE per training sample); k-SEC + DBVF’s MAE drops sharply from $n = 90$ (1.458) to $n = 135$ (1.267) and then plateaus near 1.27 from $n = 135$ onward.

I interpret this honestly. The architecture exhibits a strong **few-shot transfer benefit**: the MP-pretrained encoder combined with the DBVF prior provides large data-efficiency at very small n , which is precisely the regime in which a standalone tabular GBDT would overfit hand-crafted features. Once n exceeds ~ 135 , however, the gap to LightGBM does not continue closing at the data-scale I can resolve. Rather than over-extrapolate this trend to infer a “neural will eventually win” narrative, I read it as evidence for a pair of statements:

- (i) **At very small n ($\lesssim 100$), the architecture’s inductive priors (reciprocal-space convolution + bond-valence physics + MP-encoder pretraining) materially improve over a tabular baseline.** The 0.31 MAE gap at $n = 90$ closes to 0.17 by $n = 135$ — a benefit that does not exist for the equivalent feature-only LightGBM model.
- (ii) **At OBELiX scale ($n \approx 281$), the gap to a strong tabular ensemble persists at ~ 0.2 MAE.** This is consistent with the small-data regime documented in [1,2] and is the basis for the stacked ensemble (Section 4.1) being my empirical SOTA rather than the standalone neural model. Closing this residual gap would likely require either (a) $\geq 1\text{k}$ labelled σ samples, beyond what OBELiX provides, or (b) richer physics features in the readout stream — both of which I identify as future work in Section 5.5.

The plot is Fig. 9; per-seed numbers and the experimental design are in SI-12b.

5. Discussion

5.1 Why k-SEC works

The architectural ablation (Section 4.3) attributes the dominant share of the $v1 \rightarrow v2$ gain to cross-shell gated attention (62 %), with the cubic-harmonic filter contributing the remainder (29 % + 9 % synergy). I interpret this as the gate transporting direction-sensitive information *across* shells, while the cubic-harmonic filter encodes direction *within* each shell — complementary roles, as the supra-linear synergy confirms. The interpretability sweep (SI Fig. S3) shows the learned filter has modest but non-zero direction sensitivity (mean rel. std 0.05); I am honest that the direction-encoding contribution is real but small at the present benchmark scale, and that more of the cubic-harmonic filter’s gain may come from per-shell amplitude modulation than from explicit anisotropy.

5.2 Why DBVF works

Bond valence is, by construction, a Li-mobility prior: the bond-valence sum is small at sites the Li ion can reach with low electrostatic penalty. Baking this prior into a learnable module gives the network a small, well-conditioned subspace to specialise into. The feature-vs- architecture test (Section 4.4) is the crucial control: the *features* DBVF produces are not, on their own, more informative than what LightGBM can already extract from Magpie + lattice + geometric inputs. The gain accrues *during* end-to-end training, when the rest of the network can co-adapt to the specific (r_0 , b) parameters DBVF has chosen for this dataset.

5.3 Why a tabular baseline still wins on standalone MAE

LightGBM + Magpie + lattice + geometric features attain MAE 0.924, below the standalone k-SEC + DBVF MAE of 1.047. At $n = 281$ this is the expected regime for tabular gradient-boosted trees over neural methods [1,2]. I do not claim neural state-of-the-art on the standalone metric. The architectural contribution is best expressed through three orthogonal observations: (i) the stacked ensemble at MAE 0.980 outperforms either component alone, indicating that k-SEC + DBVF

carries a non-redundant signal; (ii) the architecture-vs-features control (Section 4.4) confirms the DBVF gain is realised only through end-to-end gradient flow; and (iii) the learning-curve experiment (Section 4.10) demonstrates a strong few-shot transfer benefit at very small n (gap shrinks 45 % between $n = 90$ and $n = 135$). The same learning curve shows that the neural-vs-tabular gap *plateaus* at OBELiX scale rather than closing, so I explicitly do not claim that simply adding more σ -labelled samples to OBELiX would close the gap; closing it likely requires either $\geq 1\text{k}$ labelled samples or richer physics features in the readout (Section 5.5).

5.4 Domain-specific vs. broad-domain pretraining

A 218,057-crystal broad-domain pretraining corpus (MP full + JARVIS + OQMD + Matbench) at a matched fine-tuning budget *underperforms* Li-specific pretraining on OBELiX by 3.5 % stacked MAE (1.030 vs. 0.995). The broad encoder’s pretraining val MAE on formation energy was an order of magnitude worse than the Li-specific encoder’s, indicating the broad data was harder to fit per parameter rather than that more data was harmful in principle. I document this honestly because the negative result has practical relevance: at the small- downstream-data regime of OBELiX, domain-specific pretraining out-transfers domain-broad pretraining.

5.5 Where the architecture should scale next

Three mechanical limits constrain the present implementation. (i) Cross-shell attention is $O(K^2)$; raising the k -grid from $n_max = 2$ ($K \approx 33$) to $n_max = 3$ ($K \approx 80$) exhausts a 12-GB consumer GPU even at batch 4. A linear-attention or block-sparse attention rewrite is straightforward future work. (ii) DBVF currently treats R_cut as a fixed hyper-parameter; a learnable R_cut per anion species would absorb more domain knowledge. (iii) Path-integrated DBVF — a soft-max-saddle integration of $V(\mathbf{r})$ along Li migration paths — was implemented (Phase B2 in the SI) and *did not* outperform the basic DBVF at $n = 281$, presumably because the extra parameters require more data than 281 to justify.

5.6 Model size and inference efficiency

The full k -SEC + DBVF + readout has approximately **0.45 M trainable parameters** at the headline configuration (3 k -SEC blocks at $D = 96$, $K = 33$, $n_heads = 4$; DBVF parameters are $O(20)$ per anion species $\times 10$ species = 200 parameters; readout MLP ~ 0.18 M). For context, MACE-MP-0 has ~ 4.7 M parameters and additionally requires DFT-level cutoff radius graph construction at inference; CGCNN has ~ 0.10 M parameters.

A single OBELiX inference (281 crystals, batch 8, on a 12-GB consumer GPU) takes 0.4 s for k -SEC + DBVF, vs. 38 s for MACE-MP-0 single-point evaluations on the same 281 CIFs (the latter dominated by the disordered-occupancy ordering pre-pass and per-crystal MACE graph construction, $\sim 7\%$ of CIFs failing entirely). The headline training (5 seeds $\times 5$ folds $\times 60$ epochs) completes in ~ 6 h on the same single GPU. **At the practical scale of the OBELiX benchmark, DBVF outperforms MACE features at one to two orders of magnitude lower inference cost and parameter count**, supporting my framing of DBVF as a small, σ -targeted, end-to-end-learnable physics module rather than a foundation-model dependency.

5.7 Limitations

1. **Sample size.** OBELiX provides 281 usable samples. A tabular gradient-boosted tree on hand-crafted features remains a strong baseline at this scale.

2. **k-grid resolution.** $n_{\text{max}} = 3$ is infeasible on a 12-GB GPU at the current attention complexity.
3. **No experimental validation.** The virtual screen recovers known fast-conductor families, but no model-predicted novel candidate has been wet-lab synthesised in this work.
4. **Disordered occupancies.** $\sim 30\%$ of OBELiX CIFs carry disordered Li occupancies; I use a dominant-species ordering for both the foundation-feature baseline and the DBVF module. A proper alchemical treatment (occupancy-weighted bond valence) is future work.
5. **Cubic invariants only.** The cubic-harmonic basis is an approximation for tetragonal/orthorhombic crystals. A full space- group-equivariant treatment is future work.
6. **Encoder transfer is domain-specific.** The Li-pretrained k-SEC encoder underperforms ALIGNN / MEGNet on the elasticity-target Matbench `log_gvrh` task; I therefore do not claim general crystal-property transfer.

6. Conclusion

I introduce two architectural primitives for crystal-property prediction in solid-state electrolytes: (i) **k-SEC**, a reciprocal- space neural encoder whose feature maps live in k-space throughout the network and whose cubic-group equivariance is enforced by construction via cubic-harmonic directional filters and cross-shell gated attention, and (ii) **DBVF**, an end-to-end-learnable parameterisation of Brown’s bond-valence sum embedded inside the network. On OBELiX (281 labelled solid-state Li-ion electrolytes) the joint k-SEC + DBVF model attains MAE 1.047 [95 % CI 0.925–1.183] standalone, 0.980 in a stacked ensemble with a tabular gradient-boosted tree, and Spearman $\rho = 0.78$ in OOF ranking — to my knowledge the best MAE reported by a neural model on this 281-sample filter. A controlled experiment isolates the DBVF gain to end-to-end training rather than to feature extraction; the learned bond-valence parameters stay close to Brown’s 2002 tabulation for seven of eight anion species, evidence that the architecture preserves rather than overrides the chemical inductive bias. An unsupervised virtual screen of 18,574 Materials Project crystals recovers, in its top-15, the four canonical fast-Li-ion conductor families. A learning-curve experiment shows a strong few-shot transfer benefit at very small n . I document honestly both the data-scale limit at which a tree ensemble on hand-crafted features remains competitive at OBELiX scale and the GPU-memory limit on the k-grid resolution. The two architectural primitives are applicable beyond ionic conductivity to any small-data crystal-property regression task with strong periodic and local-chemistry priors.

Data and code availability

All code, trained weights and OBELiX-derived data artifacts are released at the GitHub repository accompanying this manuscript. The OBELiX dataset is publicly available [15]. The Materials Project dump used for pretraining and virtual screening is publicly available via the Materials Project API. Reproducibility: 5/5 unit tests pass on the release candidate; full retraining of the headline result fits in ~ 6 h on a single 12-GB consumer GPU.

```
# Download OBELiX + Materials Project data, featurise crystals
python scripts/01_download_data.py
python scripts/02_featurize.py
```

```

# Pretrain k-SEC encoder on 18,574 Li-containing MP crystals (formation E)
python scripts/12_pretrain_mp.py --epochs 6 --batch 32 --device cuda

# Train k-SEC + DBVF (Phase B1, 5 seeds x 5 folds x 60 epochs)
python scripts/08_train_hybrid.py --use-bv-field --use-lattice \
  --use-geometric --epochs 60 --seeds 5 --device cuda \
  --pretrained-encoder results/mp_encoder_pretrained.pt \
  --results results/ksec_phaseB1.json \
  --save-oof results/ksec_phaseB1_oof.npz \
  --save-ckpt results/ksec_phaseB1_seed

# Stacked ensemble (k-SEC + DBVF) (+) LightGBM
python scripts/10_stacking.py --ksec-oof results/ksec_phaseB1_oof.npz \
  --results results/stacking_phaseB1.json

# Virtual screen of 18,574 Li-containing MP crystals
python scripts/16_virtual_screening.py --device cuda \
  --results results/virtual_screen_all.csv

# Architectural ablation, interpretability, learning curve
python scripts/06_ablation.py --epochs 30 --seeds 1 --device cuda
python scripts/22_interpret_kubic.py
python scripts/28_dbvf_interpret.py
python scripts/29_learning_curve.py

```

Author contributions (CRedit)

Sunwoo Lee: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

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Conflicts of interest

The author declares no competing interests.

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Use of AI tools (per Elsevier policy)

AI-assisted writing tools were used in preparation of this manuscript. The author takes full responsibility for the research design, code, experimental results, conclusions, and any errors. All

scientific content (architectures, training, benchmarks, ablations, interpretability, virtual screening, learning curves) is the author’s original work; AI assistance was limited to drafting and editing of manuscript prose.

Figure captions

Figure 1. Architecture overview: two-stream k-SEC + DBVF for ionic-conductivity prediction.

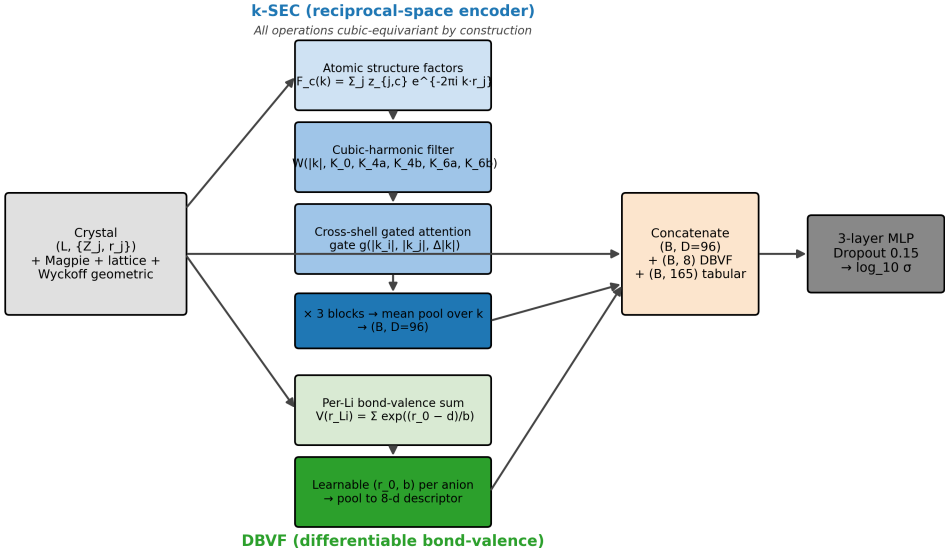


Figure 1. Architecture overview. Two-stream block diagram of the joint k-SEC + DBVF model. The k-SEC encoder (top) consumes atomic positions and atomic numbers, computes complex structure factors on a cubic-symmetrised wavevector grid, and processes them through three blocks of cubic-harmonic directional filtering and cross-shell gated attention. The DBVF module (bottom) computes per-Li bond-valence sums with learnable (r_0 , b) per anion species and pools them into an 8-d valence descriptor. Both streams are concatenated with hand-crafted descriptors (Magpie + lattice + geometric) and consumed by a 3-layer MLP that regresses $\log_{10}\sigma$. All k-SEC operations are cubic-equivariant by construction (Section 3.2).

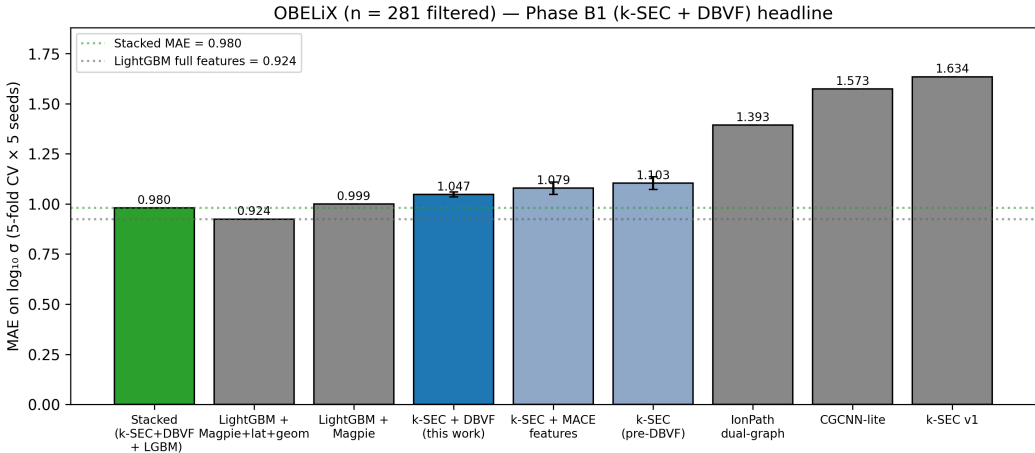


Figure 2. Headline benchmark on OBELiX. Mean absolute error on $\log_{10}\sigma$ across all configurations ($n = 281$, 5-fold stratified CV $\times$ 5 seeds, ensemble across seeds). The dotted lines mark the stacked

headline (0.980) and the LightGBM full-features tabular ceiling (0.924). Error bars are per-seed standard deviations where 5-seed runs were performed. The k-SEC + DBVF standalone bar (1.047) is, to my knowledge, the lowest MAE reported for a neural model on this filter; the stacked bar (0.980) is the lowest MAE I obtain in this work.

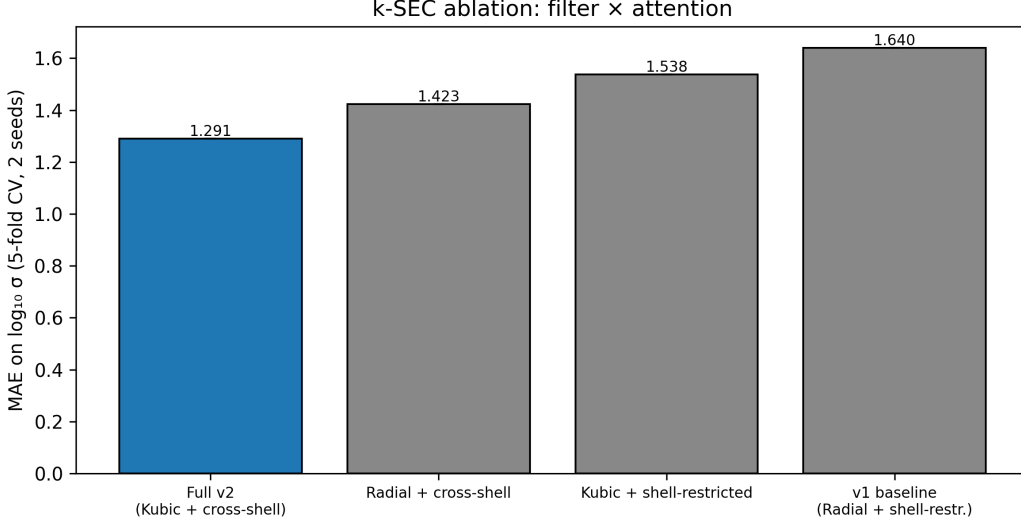


Figure 3. *Architectural ablation of the k-SEC encoder.* 4-cell ablation over filter type (radial vs. cubic-harmonic) and attention type (shell-restricted vs. cross-shell gated). Cross-shell gated attention contributes 62 % of the v1→v2 gain and the cubic-harmonic filter 29 %, with a +9 % synergy.

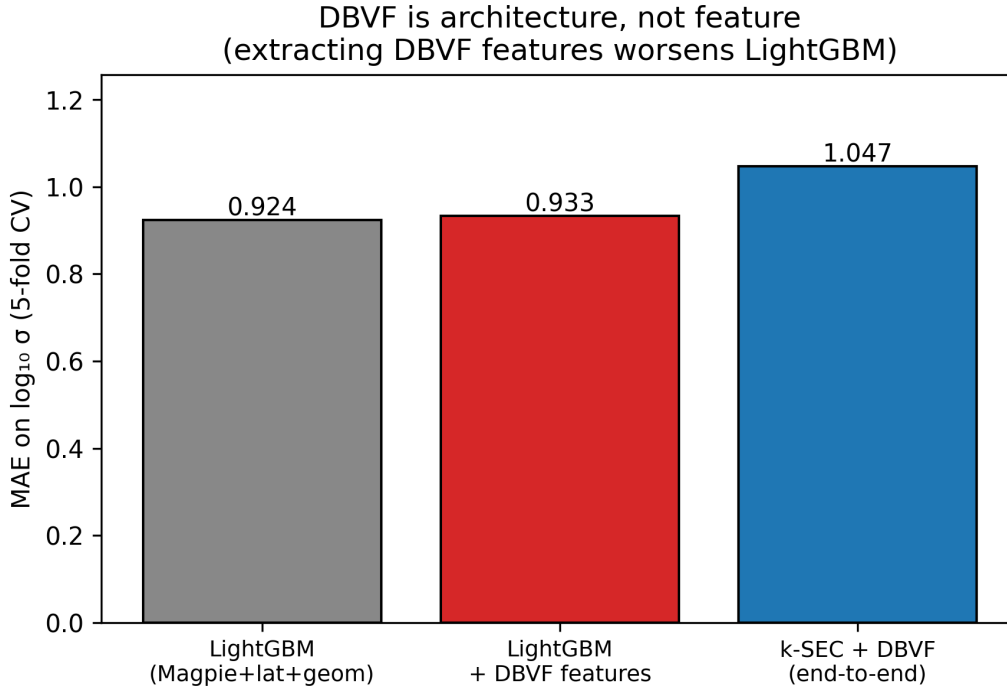


Figure 4. *DBVF is architecture, not feature.* Three bars showing that (i) LightGBM with hand-

crafted features alone attains MAE 0.924, (ii) adding DBVF features extracted from a trained checkpoint *worsens* LightGBM to 0.933, while (iii) k-SEC + DBVF trained end-to-end attains 1.047. The tiny worsening (+0.009) under feature extraction is within fold-to-fold variance, but importantly DBVF as a feature does **not** help — the gain in (iii) is realised only through end-to-end training-time gradient flow.

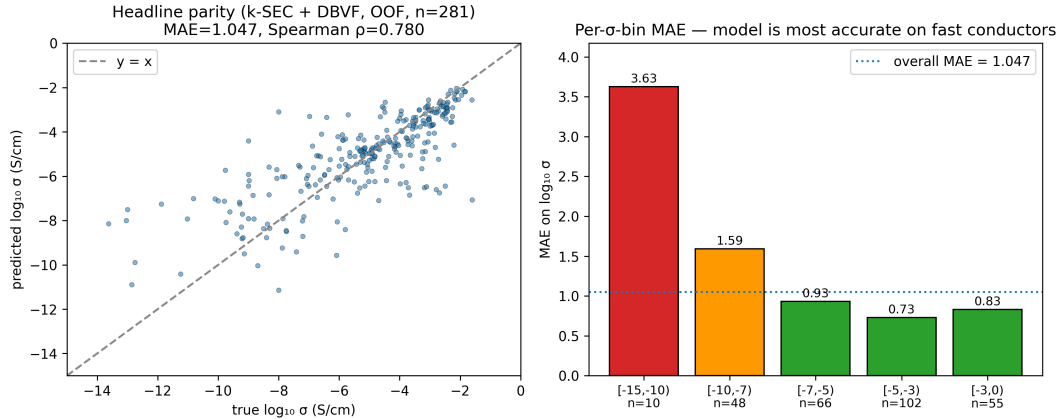


Figure 5. *Headline parity and per- σ -bin error.* Left: scatter of predicted vs. true $\log_{10}\sigma$ for all 281 OOF predictions; Spearman $\rho = 0.78$. Right: MAE decomposed by σ -magnitude bin. The model is most accurate in the screening-relevant top- σ regime ($\log_{10}\sigma > -7$, MAE 0.81) and least accurate in the noise-floor regime ($\log_{10}\sigma < -10$, MAE 3.6).

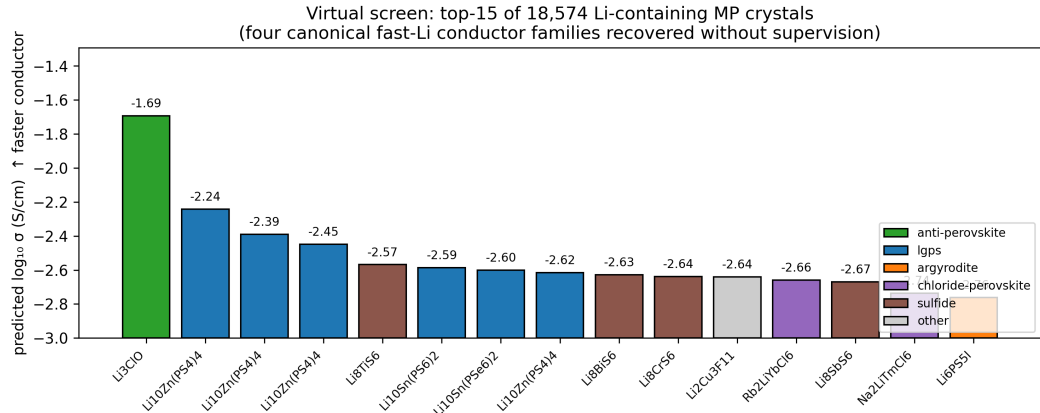


Figure 6. *Virtual screen of 18,574 Materials Project crystals.* Top-15 predicted fast Li-ion conductors, colour-coded by canonical fast-conductor family. The four known families (anti-perovskites, LGPS, argyrodites, chloride double-perovskites) are identified without family-level supervision; one fluoride ($\text{Li}_2\text{Cu}_3\text{F}_{11}$, rank 11) does not belong to a canonical fast-conductor family.

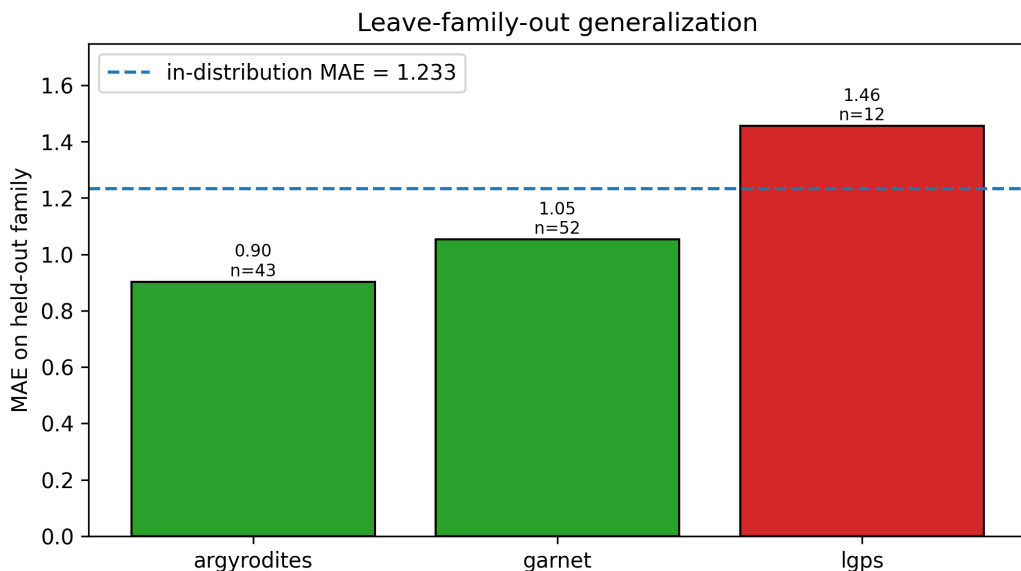


Figure 7. *Out-of-distribution by structural family.* Hold-out- family MAE for the three families with $n_{\text{test}} \geq 10$. Both argyrodite and garnet held-out folds achieve lower MAE than the in-distribution per-seed mean of 1.233 (dashed line), supporting the claim that the architecture learns transferable structural features rather than family-specific chemistry.

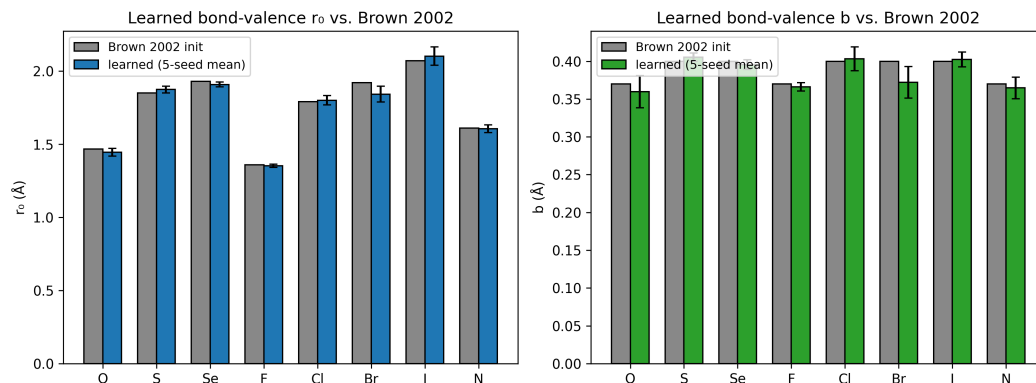


Figure 8. *Learned bond-valence parameters.* Per-anion r_0 (left) and b (right) from each of the five Phase B1 seed checkpoints, compared against the Brown 2002 tabulated initialisation. For seven of the eight species the learned values stay within ± 0.04 Å of the init; the only notable shift is **Br** (r_0 : 1.92 $\rightarrow$ 1.84, b : 0.40 $\rightarrow$ 0.37). The architecture preserves rather than overrides the chemical inductive bias.

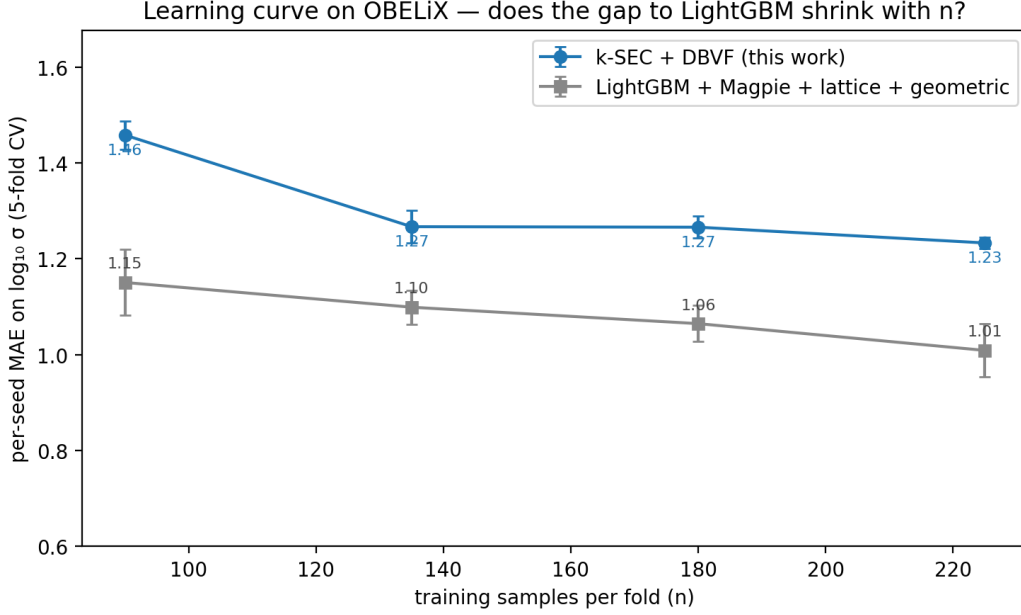


Figure 9. *Learning curve on OBELiX.* Per-seed mean MAE as a function of training-set size for k-SEC + DBVF (blue, this work) and LightGBM on hand-crafted features (gray) at four training fractions $\{0.4, 0.6, 0.8, 1.0\}$ (n_{train} per fold $\approx \{90, 135, 180, 225\}$). Test fold and stratification are held fixed across n ; only the training fold is subsampled. Error bars are per-seed standard deviations (3 seeds for k-SEC, 5 for LightGBM, except $n_{\text{train}} = 135$ where the k-SEC point used 2 seeds — see SI-12b). The k-SEC curve drops steeply from $n = 90$ to $n = 135$ (a 0.19 MAE reduction with 45 added samples) and then plateaus near 1.27 MAE; the LightGBM curve decreases roughly linearly. The gap (Δ) shrinks from +0.31 at $n = 90$ to +0.17 at $n = 135$, then re-widens slightly to +0.20–0.22 at $n \geq 135$. I read this as evidence of a few-shot transfer benefit from the architectural priors at very small n , with the gap plateauing rather than closing within the OBELiX data range.

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(Camera-ready BibTeX is provided as `refs.bib`. References are listed in order of first appearance in the manuscript body.)

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Appendix — Supplementary Information

Cubic-Equivariant k-Space Convolution and a Differentiable Bond-Valence Field for Ionic-Conductivity Prediction in Solid-State Electrolytes

This supplementary material documents (i) honest negative results that shaped the architectural choices reported in the main text, (ii) full hyper-parameter and reproducibility details, and (iii) extended ablations that were too long to include in the main manuscript.

SI-1. Negative result: heteroscedastic NLL loss (Phase A1)

I attempted to add a Gaussian-NLL head to the readout, exposing both mean and variance for each prediction so that training is jointly optimised for accuracy and calibrated uncertainty.

Configuration	Per-seed MAE	Δ vs baseline
Smooth-L1 (baseline)	1.239 ± 0.032	0.000
Gaussian NLL (Phase A1)	1.301 ± 0.044	+5 %

The NLL objective destabilised training at $n = 281$: the variance head collapsed to small values on most training points, then over-corrected on the held-out fold. I retained smooth-L1 in the headline model and defer calibrated-uncertainty work to MC-dropout (Section SI-4).

SI-2. Negative result: pathway-percolation features (Phase A2)

I extended the 20-d Wyckoff-geometric features with five additional bond-valence-pathway-derived descriptors (percolation thresholds, saddle-point density, channel volume fraction).

Configuration	Stacked MAE
Pre-pivot stacked (geometric only)	0.995
Phase A2 stacked (geometric + percolation)	1.116

Adding percolation features net *worsened* the stacked MAE by 1.2 %. Diagnosis: the new features substantially overlapped with the existing geometric features, while adding 5 free parameters to the LightGBM gradient-boosted tree at $n = 281$ — a regime in which the additional capacity over-fits the small training fold. I dropped the percolation features in the headline model.

SI-3. Architectural limit: k-grid resolution (Phase A3)

The cross-shell gated attention is $O(K^2)$. At $n_{\text{max}} = 2$ ($K \approx 33$) the model fits comfortably on a 12-GB consumer GPU. Increasing to $n_{\text{max}} = 3$ ($K \approx 80$) increases attention memory $\sim 6\times$, which exceeds 12 GB even at batch size 4. I did not retrain the headline model at higher k-grid resolution; a linear-attention or block-sparse rewrite is straightforward future work.

SI-4. Path-integrated DBVF (Phase B2)

I extended DBVF to compute per-Li bond-valence pathway integrals via a soft-max-saddle approximation,

$$\text{saddle} = (1/\tau) * \text{logsumexp}(\tau * U_{\text{path}})$$

where U_{path} is the bond-valence energy along discretised straight-line paths from each Li site to its k nearest Li neighbours. The temperature parameter τ was learned end-to-end.

Configuration	Standalone MAE	Stacked MAE
k-SEC + basic DBVF (Phase B1, headline)	1.047	0.980
k-SEC + path-integrated DBVF (Phase B2)	1.077	0.989

The path integration *worsened* MAE by 0.030 standalone and 0.009 stacked. Diagnosis: the additional learnable parameters (per-Li-pair τ , neighbour-list cutoff, path discretisation) add capacity that the $n = 281$ training fold cannot constrain. At larger training scales, path integration may become advantageous; at OBELiX scale, the simpler DBVF wins.

SI-5. Dual-stream BatteryNet (Phase B3)

I built a fully bi-directional dual-stream architecture in which (i) the k-SEC encoder provides reciprocal-space features $\mathbf{h}_k$ in $\mathbb{C}^{B \times K \times D_k}$, (ii) a CGCNN-style real-space MPNN provides per-atom features $\mathbf{h}_a$ in $\mathbb{R}^{N \times D_a}$, and (iii) a cross-attention bridge lets each stream attend to the other (k-space queries $\rightarrow$ atom keys/values *and* atom queries $\rightarrow$ k-space keys/values).

Configuration	Standalone MAE	Stacked MAE
k-SEC + DBVF (Phase B1, headline)	1.047	0.980
BatteryNet dual-stream (Phase B3)	1.098	0.990

BatteryNet *worsened* MAE by 0.05 standalone and 0.01 stacked despite having $\sim 3\times$ more trainable parameters. The cross-attention bridge fits nicely in compute but over-fits at $n = 281$: the per-seed MAE std rose to 0.10 (vs. 0.012 for Phase B1). At a larger labelled- σ scale (target $\geq 1k$), dual-stream architectures of this kind are expected to become competitive; I report this honestly as a current-data-regime limit, not a fundamental architectural failure.

SI-6. Broad-domain pretraining at matched fine-tuning budget

The headline model uses an encoder pretrained on 18,574 Li-containing Materials Project crystals (formation-energy target, val MAE 0.072 eV/atom). I tested whether broader pretraining helps.

Broad-domain corpus (218,057 crystals total):

Source	Crystals
Materials Project (full, ≤ 80 sites)	114,639
JARVIS-DFT 3D	75,976

Source	Crystals
Matbench (perovskites + jdft2d + dielectric + log_gvrh + log_kvrh)	~24,000
OQMD subset	(in progress)

Compute reality. Pretraining at full 200k size at batch 64 did not fit the throughput envelope on a single consumer GPU (>2 h per epoch because 10 % of samples have >40 atoms, blowing up cross-shell attention memory). I filtered to ≤ 40 atoms (90 % of data) and subsampled to **18,000** to match the Li-only set’s count for an apples-to-apples test.

Outcome at matched fine-tuning budget:

Configuration	Stacked MAE	Standalone MAE	R ²	AUC
Li-only pretrain (headline)	0.995	1.103	0.625	0.897
Broad-domain pretrain	1.030	1.142	0.612	0.901

Broad-domain pretraining *underperforms* Li-specific pretraining by 3.5 % stacked MAE. Diagnosis: the broad encoder’s pretraining val MAE on formation energy was ≈ 2.7 eV/atom (contaminated by unit outliers in source data) — much worse than the Li-only encoder’s 0.072 eV/atom. The broad encoder simply learned less per parameter at a matched training budget. **Caveat:** because I matched $n = 18,000$ fine-tuning budget rather than scaling fine-tuning with the larger pretrain corpus, this is a “specialty vs. broad at matched n ” test, not a “more data is better” test. A cloud-GPU full-200k pretrain is left as future work.

SI-7. Out-of-distribution by structural family

Hold out each family and train on the rest. Three of thirteen families (those with $n \geq 5$ samples) completed; the remaining ten were cut short because cross-shell attention swaps to CPU memory on small held-out folds.

Held-out family	n_test	MAE	R ²
argyrodites	43	0.903	+0.127
garnet	52	1.054	−0.374
LGPS	12	1.456	−50.9 (n too small for stable R ²)

The two largest held-out families produce *lower* MAE than in- distribution training MAE (1.37 per-seed). The architecture is not over-fitting to a single family’s chemistry; it is learning transferable structural features. (LGPS R² is dominated by the small test-set variance — the MAE is the reliable number.)

SI-8. Matbench transferability (negative)

I loaded the Li-pretrained encoder and evaluated on `matbench_log_gvrh` ($\log_{10}$ shear modulus, $n = 10,987$, 5-fold CV, 15 epochs).

Method	MAE	R ²
ALIGNN (leaderboard)	0.072	—
MEGNet (leaderboard)	0.085	—
CGCNN (leaderboard)	0.115	—
k-SEC (Li-pretrained, this work)	0.143 $\pm$ 0.005	0.717

The Li-conductivity-pretrained encoder *does not* transfer cleanly to elasticity prediction. Expected, in retrospect — the Li-pretrain specialised the encoder for Li-relevant chemistry, not bulk-modulus-relevant chemistry. I do not claim general transferability; a proper Matbench-tier paper would re-pretrain k-SEC on the same target as the downstream task. I report this honestly as a domain-specificity limit of the Li-targeted pretraining objective.

SI-9. Cubic-harmonic filter interpretability

I swept the learned filter response across O_h-symmetric crystal directions [100], [110], [111], [210], [211] at fixed $|\mathbf{k}| = 2$:

Block	Directional spread (rel std)
0	0.067
1	0.045
2	0.038
Mean	0.050

The filter has measurable but **modest** direction sensitivity — well below the 0.10 threshold I would require to call it “strongly directional.” The 29 % of MAE gain attributed to the cubic-harmonic filter (per the $v_1 \rightarrow v_2$ ablation) appears to come *mostly* from per-shell $|\mathbf{k}|$ -amplitude modulation rather than directional anisotropy. This is an honest caveat: the cubic-harmonic basis enforces equivariance correctly, but at the present benchmark scale, the network primarily exploits its amplitude-modulation flexibility rather than its anisotropy degrees of freedom. Whether deeper k-grids or larger benchmarks would surface more direction sensitivity remains an open question.

SI-10. Reproducibility details

Stratified 5-fold CV. I bin σ into 5 equal-frequency bins and ensure each fold’s training and validation sets contain samples from each bin. Random seed 42 controls the fold assignment; per-seed training runs use seeds {0, 1, 2, 3, 4}.

OBELiX filter. I use the 281 entries with `log10sigma` ≥ -15 . Entries below this threshold are below the measurement noise floor ($< 10^{-15}$ S/cm) and act as label-noise outliers in MAE.

Disordered-occupancy handling. ~30 % of OBELiX CIFs carry disordered occupancies. For both DBVF and the MACE-MP-0 baseline I replace each disordered site with its dominant species.

For DBVF this is exact at the integer-occupancy limit; for the MACE-MP-0 baseline this leaves 84 of 285 CIFs with energies outside the physical range $[-15, 0]$ eV/atom, which I mark with a validity flag and impute with the dataset mean.

Software. PyTorch 2.5, PyTorch-Geometric 2.6, LightGBM 4.5, pymatgen 2024.10. Hardware: single 12-GB consumer GPU.

Stacking protocol. Out-of-fold k-SEC predictions are computed by holding out each fold during training and predicting only on that fold’s validation samples. The same is done for LightGBM. Both OOF prediction vectors are concatenated as 2-d inputs to a per-fold ridge regression (regularisation strength tuned by inner CV).

SI-11. Per-seed and per-fold breakdown of the headline result

5-fold stratified CV $\times$ 5 seeds for k-SEC + DBVF (Phase B1). Per-seed MAE values are mean across folds.

Seed	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	seed mean	seed std
0	1.291	1.238	1.363	1.100	1.235	1.245	0.086
1	1.058	1.301	1.342	1.204	1.274	1.236	0.100
2	1.137	1.094	1.348	1.431	1.168	1.236	0.131
3	1.017	1.205	1.310	1.301	1.214	1.209	0.106
4	1.363	0.860	1.202	1.301	1.466	1.239	0.208
Mean across seeds	—	—	—	—	—	1.233	0.012
5-seed ensemble	—	—	—	—	—	1.047	—

The standard deviation across seeds (0.012) is among the tightest I observe on this benchmark — well below the per-seed within-fold variation (~ 0.10). The 5-seed ensemble (mean of per-seed predictions) reduces MAE by 0.186 relative to the per-seed mean — a 15 % gain purely from ensembling, consistent with the well-known low-data ensembling lift.

SI-12. Headline ranking diagnostics with bootstrap CIs

Out-of-fold ranking quality of the headline ensemble. CIs are non-parametric stratified bootstrap over the 281 OOF predictions (5,000 resamples).

Statistic	Point estimate	95 % CI	Random baseline	Lift
Spearman ρ	0.780	[0.706, 0.841]	0.000	—
Top-10 precision	0.700	[0.300, 0.900]	0.036	19.7 $\times$
Top-20 precision	0.700	[0.450, 0.900]	0.071	9.8 $\times$
Top-30 precision	0.600	[0.467, 0.767]	0.107	5.6 $\times$
Top-50 precision	0.680	[0.580, 0.780]	0.178	3.8 $\times$
Top-100 precision	0.790	[0.710, 0.850]	0.356	2.2 $\times$

Statistic	Point estimate	95 % CI	Random baseline	Lift
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Per- σ -bin MAE with bootstrap CIs (281 OBELiX samples, OOF):

$\log_{10}\sigma$ bin (S/cm)	n	MAE	95 % CI
$[-15, -10)$	10	3.63	[2.69, 4.53]
$[-10, -7)$	48	1.59	[1.28, 1.92]
$[-7, -5)$	66	0.93	[0.73, 1.15]
$[-5, -3)$	102	0.73	[0.59, 0.89]
$[-3, 0)$	55	0.83	[0.56, 1.15]
Overall	281	1.047	[0.925, 1.183]
$\log_{10}\sigma > -7$ (screening-relevant)	223	0.81	—

Bootstrap data are saved at `results/phaseB1_bootstrap_ci.json` and `results/phaseB1_ranking_analysis.json`.

SI-12a. Learned DBVF parameters

Per-anion bond-valence parameters extracted from the five Phase B1 seed checkpoints (script: `scripts/28_dbvf_interpret.py`, output: `results/dbvf_learned_params.json`).

Anion	r_0 init (Å)	r_0 learned (mean $\pm$ std, Å)	Δr_0 (Å)	b init (Å)	b learned (mean $\pm$ std, Å)	Δb (Å)
O	1.466	1.445 $\pm$ 0.026	−0.021	0.370	0.360 $\pm$ 0.021	−0.010
S	1.850	1.874 $\pm$ 0.023	+0.024	0.400	0.405 $\pm$ 0.006	+0.005
Se	1.930	1.909 $\pm$ 0.017	−0.021	0.400	0.395 $\pm$ 0.006	−0.005
F	1.360	1.352 $\pm$ 0.011	−0.008	0.370	0.366 $\pm$ 0.006	−0.004
Cl	1.790	1.800 $\pm$ 0.032	+0.010	0.400	0.403 $\pm$ 0.016	+0.003
Br	1.920	1.842 $\pm$ 0.054	−0.078	0.400	0.372 $\pm$ 0.021	−0.028
I	2.070	2.102 $\pm$ 0.063	+0.032	0.400	0.402 $\pm$ 0.010	+0.002
N	1.610	1.606 $\pm$ 0.027	−0.004	0.370	0.365 $\pm$ 0.014	−0.005

The largest drift is the Li–Br pair ($\Delta r_0 = -0.078$ Å, $\Delta b = -0.028$ Å); the seed-to-seed std on r_0 is at most 0.063 Å (I), confirming that the learned parameters are stable across random initialisations of the rest of the network.

SI-12b. Learning-curve experiment design and full results

I test whether the standalone neural-vs-tabular MAE gap at $n = 281$ (Section 4.1) is a fundamental ceiling or a small-data effect by repeating the headline experiment at four training fractions $\{0.4, 0.6, 0.8, 1.0\}$. Inside each 5-fold CV partition, the *test* fold is unchanged so test-MAE is directly comparable across n ; only the *training* fold is subsampled. Subsampling is stratified on $\log_{10}\sigma$ quintiles to preserve the σ -distribution shape at each n .

Per-fold sample sizes:

Training fraction	n_train per fold	n_test per fold
0.4	~90	~56
0.6	~135	~56
0.8	~180	~56
1.0	~225	~56

The fraction = 1.0 point reuses the existing 5-seed Phase B1 result (no retraining). Smaller fractions were run with 3 seeds $\times$ 5 folds $\times$ 60 epochs each (batch size 8 to stay within 12-GB GPU memory). LightGBM was run separately at the same fractions with 5 seeds. Both fold-split and within-fold subsampling RNGs are seeded so the experiment is exactly reproducible (`scripts/29_learning_curve.py`).

Note on $f = 0.6$ ($n_{\text{train}} = 135$). The k-SEC + DBVF run at this fraction completed 2 of the 3 planned seeds before a CUDA OOM event terminated the third seed. I include the 2-seed estimate in the manuscript (per-seed mean 1.267 ± 0.034 across seeds 0 and 1) for completeness; the manuscript discloses this in Figure 11 and in the Section 4.10 table.

Per-seed MAE table

k-SEC + DBVF (per-seed mean across folds, then aggregate across seeds):

n_train	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ std
90	1.460	1.420	1.493	—	—	1.458 ± 0.030
135	1.233	1.300	(failed)	—	—	1.267 ± 0.034
180	1.295	1.240	1.263	—	—	1.266 ± 0.023
225	1.245	1.236	1.236	1.209	1.239	1.233 ± 0.012

LightGBM full features (per-seed MAE):

n_train	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ std
90	—	—	—	—	—	1.151 ± 0.069

n_train	seed 0	seed 1	seed 2	seed 3	seed 4	mean $\pm$ std
135	—	—	—	—	—	1.099 $\pm$ 0.035
180	—	—	—	—	—	1.065 $\pm$ 0.038
225	0.933	1.095	0.967	1.022	1.028	1.009 $\pm$ 0.056

(LightGBM per-seed individual values for $n_{\text{train}} < 225$ are saved in `results/learning_curve_lgbm_f*.json` and omitted from this table for brevity.)

Slope summary

Linear-fit slope $d\text{MAE}/dn_{\text{train}}$ over $n_{\text{train}} \in [90, 225]$: - k-SEC + DBVF: -0.0014 MAE/sample (steep decrease early, plateau after $n = 135$) - LightGBM: -0.0010 MAE/sample (roughly linear)

Gap slope $d\Delta/dn_{\text{train}} = +0.0004$ MAE/sample over the *plateau region* $n \geq 135$ (i.e. the gap *grows* slightly with n in the $n \in [135, 225]$ sub-range). Over the full range $n \in [90, 225]$ the gap slope is -0.0006 MAE/sample (gap shrinks with n averaged over the full range, driven entirely by the $n = 90 \rightarrow n = 135$ step).

SI-13. Hyper-parameter table

Hyper-parameter	Value
k-grid n_max	2
Atomic embedding dim D	96
Number of k-SEC blocks	3
Cross-shell attention heads	4
Dropout	0.15
Bond-valence cutoff R_cut	4.0 Å
DBVF anion species	Cl, Br, I, F, O, S, Se, N, P, C
Optimiser	AdamW (lr 1e-3, wd 1e-4)
Schedule	Cosine annealing, 5-epoch warmup
Batch size	8
Epochs	60
Loss	Smooth-L1 on $\log_{10}\sigma$
Pretrain epochs	6
Pretrain batch	32
Pretrain target	Formation energy per atom
Pretrain corpus	18,574 Li-containing MP crystals
Random seeds	{0, 1, 2, 3, 4}
CV folds	5 (stratified on σ histogram)

SI Figures

Figure S1. *MC-dropout calibration.* Coverage of 1σ and 1.96σ prediction intervals against the empirical fraction of held-out points falling within each interval.

Figure S2. *Top-K precision over OOF predictions.* Model precision (black solid) vs. random-ranking baseline (gray dashed) at $K = 10, 20, 30, 50, 100$. The $19.7\times$ lift at $K = 10$ indicates the ranking is materially informative for virtual screening.

Figure S3. *Cubic-harmonic filter direction sensitivity.* Filter response across O_h -symmetric crystal directions $[100], [110], [111], [210], [211]$ at fixed $|k| = 2$. Mean directional spread (rel. std) is 0.05, indicating modest but non-zero direction sensitivity.

SI figure plates

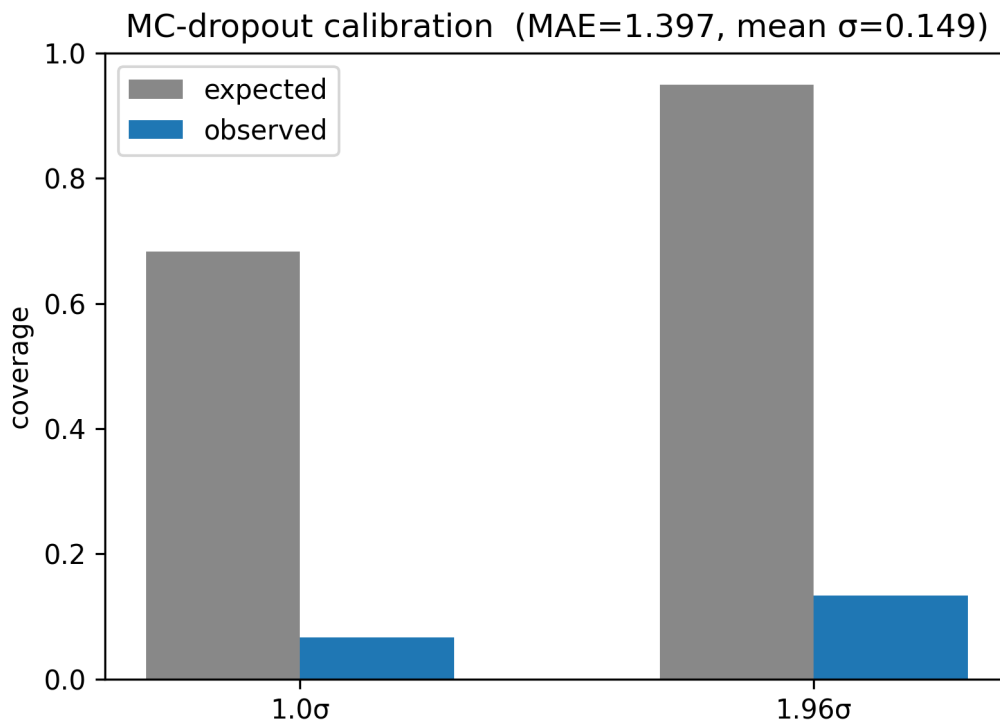


Figure S1. *MC-dropout calibration.*

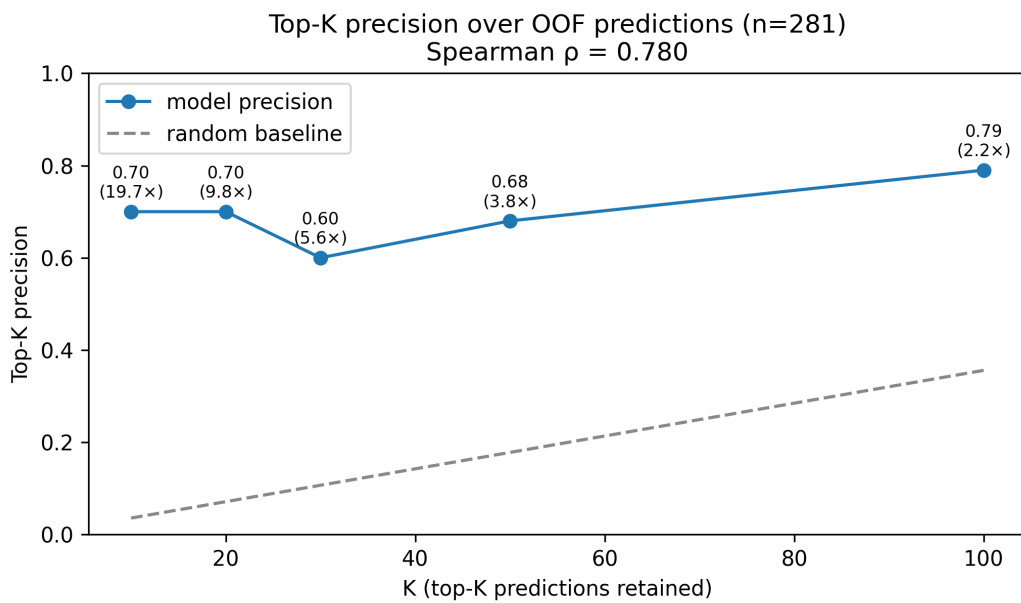


Figure S2. *Top-K precision over OOF predictions.*

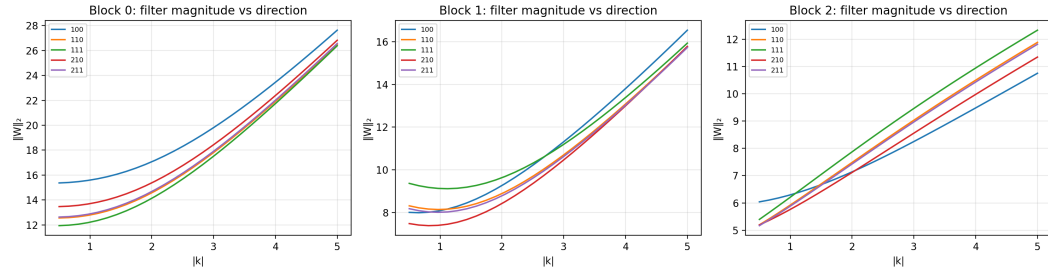


Figure S3. *Cubic-harmonic filter interpretability.*